

Grade 11 – Term 1

C6–C7 Practice Activity with Solutions

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Evaluated criterion

- C6.** Estimates interval probabilities using relative frequencies from generated samples and compares them with theoretical probabilities.
- C7.** Calculates sample means and variances for different sample sizes and compares them with the theoretical mean and variance.

1 Introduction

Problems:

- 1.1. Generated samples, sample sizes, and histograms
 - 1.2. Continuous distributions, theoretical and empirical probabilities
 - 1.3. Sample mean and variance from generated data
-

Criterion C6 focuses on estimating interval probabilities from generated samples and comparing these empirical estimates with the theoretical probabilities predicted by a continuous probability distribution.

The comparison can be understood through two representations of the same distribution. The probability density function (PDF) represents the theoretical model as a smooth curve. A histogram represents the observations obtained from a finite generated sample. The theoretical probability of an interval is determined by the area under the PDF, while the empirical probability is estimated from the relative frequency of observations in the corresponding interval.

The normal distribution provides a useful example because its theoretical density has a smooth bell-shaped form, while a histogram generated from the same distribution produces a finite and irregular approximation to that shape.

1.1 Real-world data, probability distributions, and histograms

Real-world data often show patterns of variation that can be described using probability distributions. A theoretical distribution provides a model for how values of a variable may be distributed, while a dataset contains the observations actually collected or generated from that process.

A dataset with n observations can be written as

$$x_1, x_2, \dots, x_n.$$

The observations can be organized into a histogram, which groups values into intervals and shows their frequencies. The histogram provides an empirical description of the data that can be compared with the theoretical probability distribution.

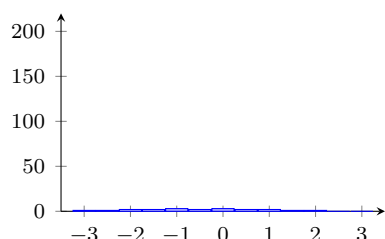


Figure 1: Count histogram from a sample of $n = 20$ observations.

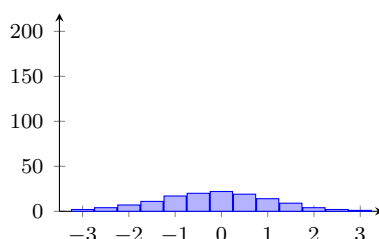


Figure 2: Count histogram from a sample of $n = 100$ observations.

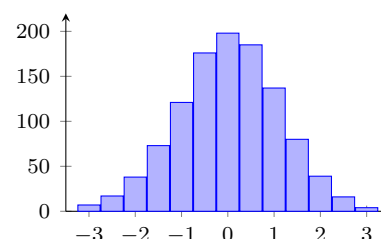


Figure 3: Count histogram from a sample of $n = 1000$ observations.

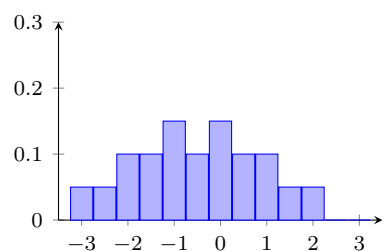


Figure 4: Histogram from a sample of $n = 20$ observations.

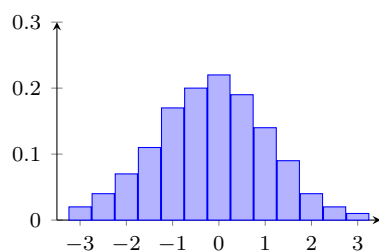


Figure 5: Histogram from a sample of $n = 100$ observations.

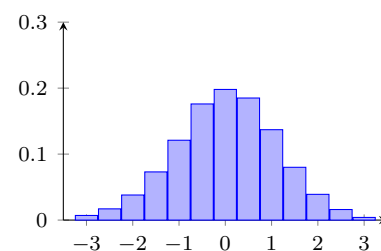


Figure 6: Histogram from a sample of $n = 1000$ observations.

Different samples from the same process can produce different histograms. Larger samples generally provide a more stable representation of the underlying distribution because random variation has less influence on the overall shape.

The theoretical distribution and the observed data can also be compared numerically. The theoretical distribution has mean μ and variance σ^2 , while the sample has mean $\bar{x}$ and variance s^2 :

Theoretical model	Observed data
Probability distribution	Observations $x_1, x_2, \dots, x_n$
Theoretical mean μ	Sample mean $\bar{x}$
Theoretical variance σ^2	Sample variance s^2

As the sample size increases, the sample statistics generally become more stable and tend to approximate the corresponding theoretical parameters:

$$\bar{x} \approx \mu, \quad s^2 \approx \sigma^2.$$

Thus, the relationship can be summarized as

Real-world process $\longrightarrow$ Observed data $\longrightarrow$ Histogram and sample statistics $\longleftrightarrow$ Probability distribution

The following subsection presents the equations used to calculate the sample mean and variance from the data.

1.2 Continuous distributions, theoretical and empirical probabilities

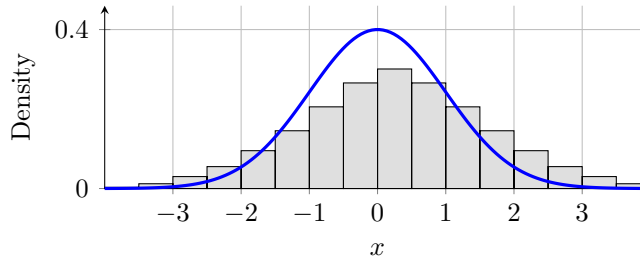


Figure 7: Theoretical normal probability density function and a histogram from a sample generated from the same distribution. The smooth curve represents the theoretical model, while the bars represent the observed sample.

Continuous probability distributions provide theoretical models for continuous random variables. For an interval $[a, b]$, the theoretical probability is determined by the area under the probability density function:

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$

When a finite sample is generated from the distribution, the corresponding probability can instead be estimated from the observations. If k of the n observations fall within the interval, the empirical probability is given by the relative frequency

$$\hat{P}(a \leq X \leq b) = \frac{k}{n}, \quad k = \#\{x_i : a \leq x_i \leq b\}.$$

The distinction between the two probabilities can be summarized as follows:

Theoretical probability	Empirical probability
Based on the probability distribution	Based on a generated sample
Area under the PDF	Relative frequency
$P(a \leq X \leq b) = \int_a^b f(x) dx$	$\hat{P}(a \leq X \leq b) = \frac{k}{n}$

The following pair of graphs illustrates the two calculations separately:

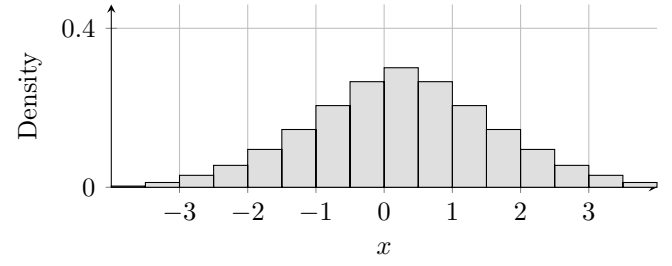
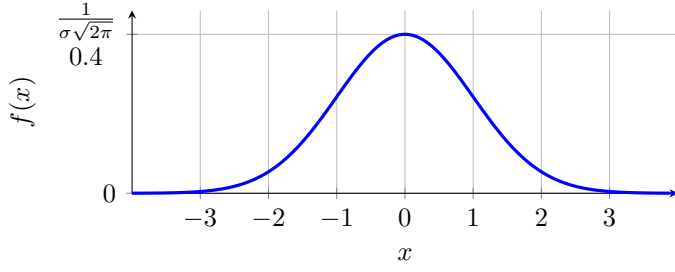


Figure 8: Theoretical probability density function of a normal distribution. The area under the curve over an interval represents the theoretical probability of that interval.

Figure 9: Histogram of a finite sample generated from the same distribution. The bars show the observed frequencies within the histogram bins.

The histogram is useful because it connects the generated observations with the empirical calculation. The bars show where the observations occur, and the observations within the selected interval determine k .

For all the continuous distributions used in this criterion, the method for calculating the empirical probability is the same:

$$\hat{P} = \frac{\text{number of observations in the interval}}{\text{total number of observations}}.$$

The method for calculating the theoretical probability depends on the probability density function of the distribution. The key comparison is therefore between the area represented by the theoretical PDF and the proportion of observations represented by the histogram:

$$\text{PDF area} \longleftrightarrow \text{histogram relative frequency}$$

For a sufficiently large generated sample, these two quantities should generally be close, while differences are expected because the histogram is based on a finite random sample.

The difference between the empirical and theoretical probabilities is

$$\text{Difference} = \hat{P}(a \leq X \leq b) - P(a \leq X \leq b).$$

The absolute difference is

$$\text{Absolute difference} = \left| \hat{P}(a \leq X \leq b) - P(a \leq X \leq b) \right|.$$

When the theoretical probability is nonzero, the percentage error is

$$\text{Percentage error} = \frac{\left| \hat{P}(a \leq X \leq b) - P(a \leq X \leq b) \right|}{P(a \leq X \leq b)} \times 100\%.$$

These quantities provide a common reference for comparing theoretical probabilities obtained from a continuous distribution with empirical probabilities obtained from generated samples.



1.3 Sample mean and variance from generated data

A theoretical probability distribution describes a model, while a finite generated sample provides observations from that model. The theoretical mean and variance are denoted by μ and σ^2 , while the corresponding sample statistics are $\bar{x}$ and s^2 .

Theoretical model	Generated sample
Probability distribution	Finite dataset
Mean μ	Sample mean $\bar{x}$
Variance σ^2	Sample variance s^2
Standard deviation σ	Sample standard deviation s

For ungrouped observations $x_1, x_2, \dots, x_n$, the sample mean, variance, and standard deviation are

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad s = \sqrt{s^2}.$$

The mean describes the center of the observed data, while the variance measures its variability around $\bar{x}$.

When observations are grouped into intervals, let f_i be the frequency of class i and m_i its midpoint:

$$m_i = \frac{a_i + b_i}{2}, \quad n = \sum_i f_i.$$

The grouped-data mean and variance are

$$\bar{x} = \frac{\sum_i f_i m_i}{n}, \quad s^2 = \frac{\sum_i f_i (m_i - \bar{x})^2}{n-1}.$$

Grouped calculations are approximate because all observations within an interval are represented by its midpoint.

Ungrouped data	Grouped data
$x_1, x_2, \dots, x_n$	Intervals and frequencies
Exact observations	Class midpoints m_i
$\bar{x} = \frac{1}{n} \sum_i x_i$	$\bar{x} = \frac{\sum_i f_i m_i}{n}$
$s^2 = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2$	$s^2 = \frac{\sum_i f_i (m_i - \bar{x})^2}{n-1}$

For a finite random sample generated from a theoretical distribution,

$$\bar{x} \approx \mu, \quad s^2 \approx \sigma^2.$$

These are approximations: different samples can produce different values. As the sample size increases, the sample statistics generally become more stable and tend toward the corresponding theoretical values.

2 C6 Practice: Empirical and Theoretical Probabilities

Problems:

- 2.1. Monthly household electricity use (raw data)
 - 2.2. Waiting times (frequency data)
 - 2.3. Package masses (grouped data)
-

The following problems focus only on estimating interval probabilities from samples. The larger samples make the relative-frequency estimates more stable, while the raw, frequency, and grouped formats provide different ways of organizing the observations.

2.1 Monthly household electricity use (raw data)

A utility company generates the following sample of monthly household electricity use (in hundreds of kWh):

3.8 4.1 4.5 5.2 5.7 6.0 6.3 6.8 7.1 7.5
4.0 4.4 4.9 5.4 5.9 6.1 6.5 6.9 7.3 8.0

For the theoretical model, $P(5 \leq X \leq 7) = 0.62$.

Question.

C6 Estimate $P(5 \leq X \leq 7)$ from the sample. Then calculate the absolute error and the percentage error relative to the theoretical probability.

C6 Solution

There are $n = 20$ observations. The values in the interval from 5 to 7, inclusive, are

5.2, 5.7, 6.0, 6.3, 6.8, 5.4, 5.9, 6.1, 6.5, 6.9,

so $k = 10$. Therefore,

$$\hat{P}(5 \leq X \leq 7) = \frac{k}{n} = \frac{10}{20} = 0.50.$$

The absolute error is

$$E = |0.50 - 0.62| = 0.12,$$

and the percentage error is

$$\frac{|0.50 - 0.62|}{0.62} \times 100\% = \frac{0.12}{0.62} \times 100\% \approx 19.35\%.$$

Thus the empirical probability is 0.50, which is 0.12 below the theoretical probability.

2.2 Waiting times (frequency data)

A clinic records the waiting times of 120 patients in the following frequency table:

Waiting time (min)	5	10	15	20	25	30
Frequency	8	20	34	30	18	10

For the theoretical model, $P(10 \leq X \leq 20) = 0.72$.

Question.

C6 Estimate the probability that a patient waits from 10 to 20 minutes, inclusive. Calculate its absolute error and compare the empirical estimate with the theoretical probability.

C6 Solution

The total frequency is

$$n = 8 + 20 + 34 + 30 + 18 + 10 = 120.$$

The waiting times 10, 15, and 20 minutes are in the interval, so

$$k = 20 + 34 + 30 = 84.$$

Hence

$$\hat{P}(10 \leq X \leq 20) = \frac{84}{120} = 0.70.$$

The absolute error is

$$E = |0.70 - 0.72| = 0.02.$$

Thus the empirical probability is 0.70, which is 0.02 below the theoretical probability of 0.72.

2.3 Package masses (grouped data)

A distribution center groups the masses of 200 packages as follows:

Package mass (kg)	0–2	2–4	4–6	6–8	8–10
Frequency	18	46	72	44	20

The classes are left-inclusive and right-exclusive. For the theoretical model, $P(2 \leq X < 8) = 0.83$.

Question.

C6 Estimate $P(2 \leq X < 8)$ from the grouped sample. Calculate the absolute error and the percentage error relative to the theoretical probability.

C6 Solution

The total frequency is

$$n = 18 + 46 + 72 + 44 + 20 = 200.$$

The interval $2 \leq X < 8$ contains the three classes 2–4, 4–6, and 6–8. Their combined frequency is

$$k = 46 + 72 + 44 = 162.$$

Therefore,

$$\hat{P}(2 \leq X < 8) = \frac{162}{200} = 0.81.$$

The absolute error is

$$E = |0.81 - 0.83| = 0.02,$$

and the percentage error is

$$\frac{|0.81 - 0.83|}{0.83} \times 100\% = \frac{0.02}{0.83} \times 100\% \approx 2.41\%.$$

Thus the empirical estimate is 0.81, which is close to, but slightly below, the theoretical probability of 0.83.

3 C7 Practice: Sample Mean, Variance, and Standard Deviation

Problems:

3.1. Machine calibration errors (raw data)

3.2. Books borrowed (frequency data)

3.3. Commute times (grouped data)

The samples in this section are deliberately small so that the sample mean, sample variance, and sample standard deviation can be calculated clearly by hand. Recall that the sample-variance denominator is $n - 1$.

3.1 Machine calibration errors (raw data)

The calibration errors (in tenths of a millimeter) from five trials are

$$4, \quad 5, \quad 6, \quad 7, \quad 8.$$

The theoretical model has $\mu = 6$ and $\sigma^2 = 2.25$.

Question.

C7 Calculate the sample mean, sample variance, and sample standard deviation. Compare the sample mean and variance with the theoretical values.

C7 Solution

There are $n = 5$ observations. First, calculate the sample mean:

$$\bar{x} = \frac{4 + 5 + 6 + 7 + 8}{5} = \frac{30}{5} = 6.$$

Next, calculate the squared deviations from $\bar{x} = 6$:

x_i	4	5	6	7	8
$x_i - \bar{x}$	-2	-1	0	1	2
$(x_i - \bar{x})^2$	4	1	0	1	4

Their sum is $4 + 1 + 0 + 1 + 4 = 10$. Therefore, the sample variance is

$$s^2 = \frac{\sum_{i=1}^5 (x_i - \bar{x})^2}{n - 1} = \frac{10}{5 - 1} = 2.50.$$

The sample standard deviation is

$$s = \sqrt{s^2} = \sqrt{2.50} \approx 1.58.$$

Thus,

$$\bar{x} = 6 = \mu, \quad s^2 = 2.50 \approx \sigma^2 = 2.25, \quad s \approx 1.58.$$

3.2 Books borrowed (frequency data)

The numbers of books borrowed by four students are summarized below:

x_i	2	4	6
f_i	1	2	1

The theoretical model has $\mu = 4$ and $\sigma^2 = 3$.

Question.

C7 Calculate the sample mean, sample variance, and sample standard deviation from the frequency table. Compare the sample mean and variance with the theoretical values.

C7 Solution

The total frequency, and hence the sample size, is

$$n = 1 + 2 + 1 = 4.$$

The sample mean is

$$\bar{x} = \frac{\sum_i f_i x_i}{n} = \frac{1(2) + 2(4) + 1(6)}{4} = \frac{16}{4} = 4.$$

Using $\bar{x} = 4$, calculate each weighted squared deviation:

x_i	2	4	6
f_i	1	2	1
$(x_i - \bar{x})^2$	4	0	4
$f_i(x_i - \bar{x})^2$	4	0	4

The sum of the weighted squared deviations is $4 + 0 + 4 = 8$. Thus,

$$s^2 = \frac{\sum_i f_i (x_i - \bar{x})^2}{n - 1} = \frac{8}{4 - 1} = \frac{8}{3} \approx 2.67.$$

The sample standard deviation is

$$s = \sqrt{s^2} = \sqrt{\frac{8}{3}} \approx 1.63.$$

Therefore,

$$\bar{x} = 4 = \mu, \quad s^2 \approx 2.67 \approx \sigma^2 = 3, \quad s \approx 1.63.$$

3.3 Commute times (grouped data)

The commute times of five employees are grouped into three classes:

Commute time (min)	0–10	10–20	20–30
Frequency	1	3	1

The theoretical model has $\mu = 15$ minutes and $\sigma^2 = 48$ minutes².

Question.

C7 Use class midpoints to estimate the sample mean, sample variance, and sample standard deviation. Compare the estimated mean and variance with the theoretical values.

C7 Solution

First, calculate each class midpoint:

$$m_1 = \frac{0 + 10}{2} = 5, \quad m_2 = \frac{10 + 20}{2} = 15, \quad m_3 = \frac{20 + 30}{2} = 25.$$

The total frequency is

$$n = 1 + 3 + 1 = 5.$$

Using the midpoints as representative values, the estimated sample mean is

$$\bar{x} = \frac{\sum_i f_i m_i}{n} = \frac{1(5) + 3(15) + 1(25)}{5} = \frac{75}{5} = 15 \text{ min.}$$

Next, calculate the weighted squared deviations:

m_i	5	15	25
f_i	1	3	1
$(m_i - \bar{x})^2$	100	0	100
$f_i(m_i - \bar{x})^2$	100	0	100

Their sum is $100 + 0 + 100 = 200$. The estimated sample variance is

$$s^2 = \frac{\sum_i f_i (m_i - \bar{x})^2}{n - 1} = \frac{200}{5 - 1} = 50 \text{ min}^2.$$

The estimated sample standard deviation is

$$s = \sqrt{s^2} = \sqrt{50} \approx 7.07 \text{ min.}$$

Therefore,

$$\bar{x} = 15 \text{ min} = \mu, \quad s^2 = 50 \text{ min}^2 \approx \sigma^2 = 48 \text{ min}^2, \quad s \approx 7.07 \text{ min.}$$

These results are estimates because all observations in a class are represented by the class midpoint.

4 Samples, Empirical Probabilities, and Sample Statistics

Problems:

4.1. Investment returns (raw data)

4.2. Daily energy consumption (frequency data)

4.3. Delivery times (grouped data)

The following examples use small samples so that each step of the calculations can be shown clearly. In real-world applications, sample sizes are typically much larger to obtain more reliable estimates.

4.1 Investment returns (raw data)

A financial analyst records these returns (in percent) over six days:

$$2, \quad 2, \quad 3, \quad 3, \quad 4, \quad 4.$$

The theoretical model has $\mu = 3$, $\sigma^2 = 1$, and $P(2 \leq X \leq 4) = 0.80$.

Question.

C6 Calculate the empirical probability of a return between 2% and 4%, inclusive, and its absolute error.

C7 Calculate the sample mean and sample variance, and compare them with the theoretical values.

C6 Solution

All six observations are in the interval, so $n = 6$ and $k = 6$. Hence

$$\hat{P}(2 \leq X \leq 4) = \frac{k}{n} = \frac{6}{6} = 1.$$

The absolute error is

$$E = |\hat{P} - P| = |1 - 0.80| = 0.20.$$

Thus the empirical probability is 1, which is 0.20 above the theoretical probability.

C7 Solution

The sample mean is

$$\bar{x} = \frac{2 + 2 + 3 + 3 + 4 + 4}{6} = \frac{18}{6} = 3.$$

Using $\bar{x} = 3$, the sample variance is

$$s^2 = \frac{(2-3)^2 + (2-3)^2 + (3-3)^2 + (3-3)^2 + (4-3)^2 + (4-3)^2}{6-1} \\ = \frac{1+1+0+0+1+1}{5} = \frac{4}{5} = 0.80.$$

Therefore,

$$\bar{x} = 3 = \mu, \quad s^2 = 0.80 \approx \sigma^2 = 1.$$

4.2 Daily energy consumption (frequency data)

A facility records daily consumption (in MWh) as follows:

x_i	8	10	12
f_i	1	2	1

The theoretical model has $\mu = 10$, $\sigma^2 = 3$, and $P(10 \leq X \leq 12) = 0.60$.

Question.

C6 Calculate the empirical probability of consumption between 10 and 12 MWh, inclusive, and its absolute error.

C7 Calculate the sample mean and sample variance, and compare them with the theoretical values.

C6 Solution

The total frequency and the frequency in the interval are

$$n = 1 + 2 + 1 = 4, \quad k = 2 + 1 = 3.$$

Therefore,

$$\hat{P}(10 \leq X \leq 12) = \frac{3}{4} = 0.75,$$

and the absolute error is

$$E = |0.75 - 0.60| = 0.15.$$

Thus the empirical probability is 0.75, which is 0.15 above the theoretical probability.

C7 Solution

The sample mean is

$$\bar{x} = \frac{1(8) + 2(10) + 1(12)}{4} = \frac{40}{4} = 10.$$

The sample variance is

$$\begin{aligned}s^2 &= \frac{1(8-10)^2 + 2(10-10)^2 + 1(12-10)^2}{4-1} \\ &= \frac{4+0+4}{3} = \frac{8}{3} \approx 2.67.\end{aligned}$$

Therefore,

$$\bar{x} = 10 = \mu, \quad s^2 \approx 2.67 \approx \sigma^2 = 3.$$

4.3 Delivery times (grouped data)

A company groups delivery times into three classes:

Delivery time (min)	10–20	20–30	30–40
Frequency	1	2	1

The theoretical model has $\mu = 25$ minutes, $\sigma^2 = 75$ minutes², and $P(20 \leq X < 30) = 0.45$.

Question.

- C6** Estimate the empirical probability of a delivery time from 20 up to (but not including) 30 minutes, and calculate its absolute error.
- C7** Estimate the sample mean and sample variance using class midpoints, and compare them with the theoretical values.

C6 Solution

The total frequency is $n = 1 + 2 + 1 = 4$. The selected class has frequency $k = 2$, so

$$\hat{P}(20 \leq X < 30) = \frac{2}{4} = 0.50.$$

The absolute error is

$$E = |0.50 - 0.45| = 0.05.$$

Thus the empirical probability is 0.50, which is 0.05 above the theoretical probability.

C7 Solution

The three class midpoints are

$$15, \quad 25, \quad 35.$$

Using these midpoints, the estimated sample mean is

$$\bar{x} = \frac{1(15) + 2(25) + 1(35)}{4} = \frac{100}{4} = 25 \text{ min.}$$

The estimated sample variance is

$$\begin{aligned} s^2 &= \frac{1(15 - 25)^2 + 2(25 - 25)^2 + 1(35 - 25)^2}{4 - 1} \\ &= \frac{100 + 0 + 100}{3} = \frac{200}{3} \approx 66.67 \text{ min}^2. \end{aligned}$$

Therefore,

$$\bar{x} = 25 \text{ min} = \mu, \quad s^2 \approx 66.67 \text{ min}^2 \approx \sigma^2 = 75 \text{ min}^2.$$

These grouped-data results are estimates because every observation in a class is represented by its midpoint.